

Notes: Baker, Jensen, and Murphy (JF, 1988)

Compensation and Incentives: Practice vs. Theory

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1 Big picture

- **Question.** Why do actual compensation and internal incentive systems look so different from the simple economic theory of optimal incentives?
- **Core observation.** Many organizations do not rely heavily on explicit pay-for-performance. Instead, they use salary grades, performance reviews with little dispersion, promotion tournaments, profit sharing, survey-based pay, tenure, and long-term employment relations.
- **Main puzzle.** Standard principal-agent theory says compensation should reward measured performance. Actual organizations often appear to mute explicit incentives, avoid punishments, avoid poor evaluations, and rely on promotions rather than transitory bonuses.
- **Central claim.** The gap between theory and practice is not a minor institutional detail. It is central to the theory of the firm, because internal incentives largely determine how people inside firms behave.
- **Perspective.** The paper does not present one fully solved model. Instead, it catalogs several robust compensation practices that are difficult to reconcile with existing economic theory and asks economists to either explain them or incorporate ideas such as fairness, morale, trust, social responsibility, and culture.
- **Economic takeaway.** Incentive systems are powerful but dangerous. Strong explicit incentives can induce people to optimize the measured task rather than the true objective. Firms may therefore choose weaker, more implicit, and more political incentive systems.

2 Incremental contribution of the paper

- **Moves agency theory inside the firm.** Earlier agency research focused on broad principal-agent contracting and corporate governance. This paper emphasizes the detailed internal incentive systems through which organizations actually manage employees.
- **Connects finance, labor, and organizational economics.** The paper links executive compensation, promotion ladders, performance evaluation, profit sharing, and efficiency wages as parts of one broader incentive problem.
- **Highlights practice-theory gaps.** Rather than forcing observed practices into existing models, the paper treats anomalies as research opportunities: weak pay-for-performance, promotion-based incentives, biased evaluations, salary surveys, lack of bonding, and reluctance to fire.
- **Emphasizes measurement problems.** The authors argue that pay-for-performance may fail not because money is irrelevant, but because performance measures are incomplete, manipulable, or politically costly to apply.
- **Introduces layered agency as a unifying idea.** In real hierarchies, the person designing or enforcing incentives is usually not the residual claimant. Supervisors and boards bear personal and political costs from tough decisions while capturing little of the value created by those decisions.
- **Influences later research.** The paper helps motivate later work on executive incentives, subjective performance evaluation, relational contracts, multitasking, internal labor markets, career concerns, corporate culture, and incentive design in organizations.

3 Structure of the argument

The paper is best read as a sequence of compensation puzzles. Each puzzle has three parts:

1. a common organizational practice;
2. the reason it is hard to explain using standard incentive theory;
3. a possible economic explanation or research direction.

Interpretation: The paper’s purpose is diagnostic rather than purely positive. It identifies where standard theory is too thin and where richer models of organizations need to include measurement costs, politics, distorted monitoring incentives, and behavioral or social constraints.

4 The benchmark: simple incentive theory

4.1 Basic economic prediction

In a simple agency model, an employee chooses effort or action, and the firm observes some signal of output. If higher effort is costly to the employee but valuable to the firm, the firm should make expected compensation increasing in measured performance.

A schematic way to write the problem is:

$$\max_{w(s)} \mathbb{E}[y(a) - w(s)]$$

subject to participation and incentive constraints:

$$\begin{aligned} \mathbb{E}[u(w(s)) - c(a)] &\geq \bar{U}, \\ a &\in \arg \max_{a'} \mathbb{E}[u(w(s)) - c(a')]. \end{aligned}$$

In this benchmark, compensation should vary with signals that are informative about the employee’s action and should not vary with signals outside the employee’s control.

4.2 Why practice is hard to explain

- Explicit bonus pay is often small compared with total compensation.
- Performance ratings are frequently compressed, with most workers receiving good ratings and very few receiving poor ratings.
- Promotions, not annual bonuses, often create the largest compensation changes.
- Salary levels are often benchmarked to job surveys, seniority, and internal equity rather than individual marginal product.
- Firms rarely require employees to post bonds or pay up-front fees for jobs, even when such contracts could theoretically solve shirking problems.

Interpretation: The observed compensation system looks less like a clean contingent contract and more like a mixture of weak explicit incentives, implicit career concerns, subjective evaluations, organizational politics, and social norms.

5 Puzzle 1: weak pay-for-performance

5.1 Observed practice

The paper argues that explicit financial rewards in the form of transitory performance-based bonuses rarely account for a large share of compensation. The authors point to evidence from Medoff and Abraham showing little compensation dispersion across performance categories for managerial and professional employees.

5.2 Why this is puzzling

If performance requires costly effort, then stronger pay-performance links should improve incentives. In a simple model, the worker's compensation should change meaningfully with observed productivity. But the observed pay premia for moving across performance ratings are small.

5.3 A better interpretation: incentives may be too effective

The authors push back against the view that money simply does not motivate. Their key interpretation is subtler:

- strong monetary incentives motivate employees to do exactly what is rewarded;
- if the measured task is not identical to the true objective, employees will distort effort toward the measure;
- therefore, strong pay-for-performance can create gaming, quality reductions, short-termism, lobbying over metrics, and manipulation of standards.

Interpretation: The problem is not that monetary incentives are weak. The problem is that they can be too powerful when the performance measure is incomplete.

5.4 Objective versus subjective performance measurement

Objective measures are easy to verify, but employees can game them. Piece rates can reduce quality, sales incentives can encourage bad sales, and accounting-profit bonuses can encourage short-term earnings management.

Subjective measures can use richer information, but they require trust and create conflict. Employees may distrust supervisors, and supervisors may dislike giving negative evaluations.

Economic message: The optimal incentive contract cannot be separated from the quality of performance measurement. If the output measure is distorted, stronger incentives may reduce value.

6 Puzzle 2: promotion-based incentives

6.1 Observed practice

Most organizations rely heavily on promotions. Large compensation increases often occur when employees move to higher job levels, not when they remain in the same job and perform marginally better.

6.2 Two roles of promotion

Promotions serve two distinct functions:

- **Matching.** Employees differ in skills, and jobs differ in requirements. Promotions help allocate people to jobs that fit their talents.
- **Incentives.** Higher ranks bring pay, prestige, and authority. The prospect of promotion motivates lower-level employees.

6.3 The tension between matching and incentives

Promotion tournaments can motivate effort, but they may promote the best performer in the current job rather than the best candidate for the next job. The classic example is that the best salesperson may not be the best sales manager.

Interpretation: Promotions cannot generally solve both problems at once. A tournament that rewards current output may create incentives, but it can also misallocate talent. A system that promotes based on expected next-job performance may improve matching, but it can weaken incentives in the current job.

6.4 Growth requirement

Promotion-based incentives require vacancies. They work more easily in growing firms and become problematic in stagnant or declining firms. The authors connect this to free cash flow and the incentive for managers to expand organizations even when growth is not value-maximizing.

Economic message: Promotion ladders create incentives for organizational growth. In low-growth environments, they may encourage value-destroying expansion or require firms to substitute toward bonuses, equity, restructuring, or stronger ownership incentives.

7 Puzzle 3: tenure and up-or-out systems

Tenure and up-or-out systems are common in universities, law firms, accounting firms, consulting, and some professional services. Employees work for several years with limited explicit pay-for-performance; then a subset is promoted to tenure or partnership, and unsuccessful candidates often leave.

- These systems seem more about **selection and matching** than about ongoing incentive provision.
- They are especially common where output is hard to evaluate quickly and where human capital, creativity, and judgment are central.
- Up-or-out rules avoid the problem of retaining disappointed employees with little future promotion prospect.
- But they also destroy organization-specific human capital if some nonpromoted employees would still be productive in lower roles.

Interpretation: Tenure and up-or-out systems solve some incentive and selection problems by making the promotion decision high-stakes. But they create other costs by forcing exit and by relying heavily on imperfect subjective evaluation.

8 Puzzle 4: profit sharing and broad-based ownership

Profit-sharing plans make worker compensation depend on overall firm or team performance. The paper notes that these plans are common and often viewed by practitioners as productivity-enhancing.

8.1 Why economists find this puzzling

In a large firm, an individual employee receives only a tiny fraction of the gains from extra effort:

$$\text{private marginal benefit} \approx \frac{1}{n} \times \text{increase in firm profit},$$

where n is the number of participants. As n grows, the direct incentive effect becomes very small.

8.2 Possible channels

- Profit sharing may encourage mutual monitoring, but this itself can suffer from free riding.
- Profit sharing may change corporate culture by making employees feel like owners.
- Profit sharing may create identification with the firm, increase communication, or support peer enforcement.

Interpretation: The puzzle is that broad-based profit sharing appears too weak to motivate individual effort through direct marginal incentives, yet firms use it widely. A richer model must explain culture, identity, peer monitoring, or group norms.

9 Puzzle 5: biased and compressed performance evaluations

9.1 Observed practice

Performance ratings often display strong compression. Very few employees receive poor evaluations, and large fractions receive favorable ratings.

9.2 Why this matters

Compressed evaluations undermine incentives because employees receive little information and little differential reward based on performance. They also make it difficult to fire or discipline weak performers.

9.3 The role of self-serving beliefs

The paper points out that many employees believe they are above average. If people overestimate their own performance, then negative evaluations create conflict, dissatisfaction, and morale costs. Supervisors may avoid these costs by inflating ratings.

9.4 Rewards versus punishments

Economically, a salary plus bonus and a salary with a penalty can be equivalent. Organizationally, they are not treated as equivalent. Firms typically frame compensation as rewards rather than penalties. This may reflect loss aversion, conflict avoidance, or the politics of giving bad news.

Economic message: Performance evaluation is not a neutral measurement technology. It is a political and psychological process. Supervisors bear personal costs from truthful negative ratings, so they may avoid them unless they are strongly incentivized to be accurate.

10 Puzzle 6: survey-based CEO pay and firm size

A major empirical regularity discussed in the paper is the relation between executive pay and firm size. Compensation surveys show that CEO salary and bonus rise with firm sales, with an elasticity around 0.3 across industries and years.

10.1 Why this is puzzling

It is plausible that larger firms need higher-quality CEOs and therefore pay more. But the puzzle is that the relation is so stable and appears to influence pay-setting directly. If compensation committees use survey regressions mechanically, then pay becomes tied to firm size rather than value creation.

10.2 Potential distortion

If CEO pay rises with firm size even when firm value does not rise, then CEOs may have incentives to expand sales, assets, or organizational scale in ways that do not maximize shareholder wealth.

Interpretation: Compensation benchmarking can transform an empirical correlation into an incentive system. If pay surveys reward size, then pay surveys may encourage empire building.

11 Puzzle 7: top-manager incentive contracts

The paper discusses executive compensation evidence showing that CEO pay-performance sensitivity is positive but very small. For example, Jensen and Murphy find that each \$1,000 change in shareholder wealth corresponds to only a tiny change in CEO salary and bonus.

Standard theory also suggests that compensation should filter out shocks outside the executive's control. This implies relative performance evaluation: pay should depend on performance relative to industry or market benchmarks. Yet many CEO incentives are based on absolute stock returns through equity ownership and stock options, not relative returns.

- Boards often do not fully adjust pay for CEO stock ownership.
- Cash pay is weakly tied to shareholder value.
- Stock options and stock ownership create stronger incentives, but usually relative to absolute price movements.
- Political and public-pressure constraints may limit very high-powered CEO pay contracts.

Economic message: Even at the top of the firm, where incentives should be easiest to motivate conceptually, actual contracts do not look like the clean contracts in theory. This suggests that political constraints, board incentives, and public perceptions matter.

12 Puzzle 8: efficiency wages and the absence of bonding

Efficiency wage models argue that paying workers more than their outside option can reduce shirking, because workers lose a valuable rent if fired. But if the present value of employment rents is positive, an efficient contract could in principle charge an up-front fee or require a bond.

The puzzle is that such up-front job fees and bonding contracts are rarely observed in ordinary employment relationships, even though they exist in some franchise settings.

12.1 Three dimensions of compensation

The authors distinguish three dimensions of compensation:

- **Level:** the expected value of the compensation package.
- **Functional form:** how compensation varies with performance.

- **Composition:** the mix of salary, bonus, benefits, workplace quality, leisure, status, and other rewards.

Interpretation: Efficiency wages are not simply about a high wage level. They are about the functional form of the employment contract: workers have something to lose if they shirk. The unresolved question is why these rents are not offset by bonds or fees.

13 A unifying mechanism: what happens when the principal is not a principal?

The final section offers one of the paper's most important economic insights. In standard principal-agent models, the principal is often assumed to be a 100 percent residual claimant. In real organizations, the people designing and enforcing incentives are usually agents themselves.

13.1 Layered agency problems

- A supervisor deciding whether to fire a weak employee bears the personal and political costs of firing.
- The same supervisor receives only a small share of the economic value created by removing the employee.
- Therefore, the supervisor may choose not to fire, not to give a poor evaluation, or not to enforce a high-powered incentive system.
- The supervisor may justify the action using language of fairness, morale, or trust, even when the underlying problem is weak incentives for the monitor.

13.2 Why this matters for compensation design

If the monitor is not the residual claimant, then optimal contracts at one layer may fail because incentives are not properly aligned at higher layers. The problem cascades upward to CEOs and boards of directors. Boards may avoid tough CEO compensation contracts or termination decisions for the same reason supervisors avoid tough employee decisions.

Economic message: The paper's deepest contribution is not merely that incentives are muted. It is that muted incentives may arise because the designers and enforcers of incentives are themselves poorly incentivized. This makes incentive design an organizational system problem, not a single contract problem.

14 Original tables and figure from the paper

14.1 Table I and Figure 1

Table I
Salary Premiums Associated With Performance Ratings, and
Frequency Distribution of Performance Ratings, for 7,629
Managers in Two Large Manufacturing Firms

Performance Rating	Salary Premium Relative to Lowest Performance Rating	Percent of Sample Receiving Performance Rating
(1)	(2)	(3)
<i>Company A (4,788 managers):</i>	%	%
Not Acceptable	-0-	.2
Acceptable	1.4	5.3
Good	5.3	74.3
Outstanding	7.8	20.2
<i>Company B (2,841 managers):</i>		
Unacceptable	-0-	-0-
Minimum Acceptable	-0-	-0-
Satisfactory	-0-	1.2
Good	1.8	36.6
Superior	3.6	58.4
Excellent	6.2	3.8

Source: Medoff and Abraham [32], Tables I and II. Salary premiums are estimated from regressions of log (earnings) on performance-rating dummies and demographic variables.

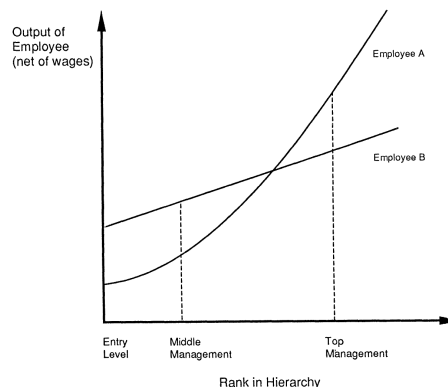


Figure 1. Relation between output (net of wages) in jobs at different levels in the hierarchy for employees of type A and B. Type-A employees, who are best suited for top management jobs, will lose promotion tournaments at lower management levels.

Table I. Salary premiums and performance ratings.

Figure 1. Promotion tournaments and the matching problem.

Read:

- **Table I** reports evidence from 7,629 managers in two large manufacturing firms. Salary premia across performance categories are small. In Company A, moving from the lowest rating to Outstanding corresponds to only a 7.8 percent premium; in Company B, moving to Excellent corresponds to only 6.2 percent.
- The distribution of ratings is highly compressed. In Company A, 74.3 percent of managers are rated Good and 20.2 percent are rated Outstanding. In Company B, 58.4 percent are rated Superior and 36.6 percent are rated Good.
- **Figure 1** shows why promotion tournaments can fail at matching. Employee B outperforms Employee A at lower ranks, but Employee A is better suited for top management. A tournament based on lower-level output promotes the wrong person.

Economic message: Table I shows that formal performance ratings and salary premia are too compressed to provide strong explicit incentives. Figure 1 shows that promotion-based incentives can motivate effort but may allocate talent poorly. Together, the two visuals capture the paper's central tension: firms need incentives, but the mechanisms they use are blunt and often conflict with other organizational goals.

14.2 Table II

Table II
Estimated Elasticity of CEO Salary and Bonus With
Respect to Firm Sales, 1973–1983

Industry	Year				
	1973	1975	1979	1981	1983
Manufacturing	.313	.296	.297	.287	.285
Retail Trade	.253	.271	.230	.306	.298
Gas & Electric Utilities	.331	.236	.347	.313	.314
Commercial Banking	.317	.329	.367	.372	.404
Insurance	.313	.277	.299	.372	.345

Source: *Top Executive Compensation*, The Conference Board, various editions. Elasticities correspond to the estimated coefficient from a regression of $\text{Log}(\text{Salary} + \text{Bonus})$ on $\text{Log}(\text{Sales})$. Sales are defined as operating revenues for utilities, deposits for banks, and total premium income for insurance companies.

Table II. Estimated elasticity of CEO salary and bonus with respect to firm sales.

Read:

- Table II reports compensation-survey estimates for five industries between 1973 and 1983.
- The elasticity of CEO salary and bonus with respect to firm sales is consistently around 0.3.
- The relation is stable across industries and years: manufacturing, retail, utilities, banking, and insurance all display similar pay-size elasticities.

Economic message: The table suggests that executive pay-setting practices may mechanically reward firm size. If compensation committees benchmark pay using survey regressions of pay on sales, then CEOs may be rewarded for growing the firm even when growth does not raise shareholder value.

15 How to use the paper in a literature review

15.1 Connection to agency theory

The paper is a foundational critique of the simple principal-agent model as a complete description of real organizations. It does not reject agency theory. Instead, it argues that agency theory must be extended to incorporate performance-measurement costs, subjective evaluation, layered agency, social constraints, and organizational politics.

15.2 Connection to executive compensation

The discussion of CEO pay-performance sensitivity and pay-size elasticity anticipates a large literature on whether executive contracts are efficient, politically constrained, or shaped by managerial power and benchmarking.

15.3 Connection to internal labor markets

The sections on promotion, tenure, up-or-out systems, and performance evaluations are closely related to internal labor-market research. The core insight is that compensation and job assignment are linked: firms cannot choose incentive systems without also choosing promotion systems, evaluation systems, and career paths.

15.4 Connection to corporate governance

The paper argues that incentive distortions inside the firm can help explain inefficient growth, weak monitoring, bureaucratic behavior, and reluctance to discipline managers. This connects internal compensation design to takeover activity, restructuring, and free-cash-flow problems.

16 Limitations and open issues

- **No unified model.** The paper identifies many puzzles but does not solve them in one formal framework.
- **Limited direct causal evidence.** Much of the empirical material is descriptive and based on examples or prior studies rather than a single research design.
- **Behavioral concepts remain under-modeled.** Ideas such as fairness, morale, trust, and culture are discussed as important, but the paper does not fully integrate them into a formal economic model.
- **Measurement remains central.** The paper argues that performance measurement is difficult, but it does not offer a complete theory of which tasks should use which incentive contracts.
- **Layered agency is suggestive.** The idea that supervisors lack incentives to enforce efficient contracts is powerful, but the paper leaves open how competition, governance, and organizational design discipline these distortions.

Interpretation: The paper should be read as a research agenda. Its importance lies in naming the puzzles clearly and showing why actual compensation practice cannot be understood by looking only at one worker and one fully residual-claimant principal.

17 Short summary

Baker, Jensen, and Murphy study the gap between economic theories of compensation and actual incentive practices in firms. Traditional theory predicts meaningful pay-for-performance, but many firms use compressed performance ratings, weak bonuses, promotion ladders, profit-sharing plans, survey-based salary systems, and long-term employment relations. The authors argue that strong incentives can be counterproductive when performance measures are incomplete and manipulable. They also emphasize that promotions must perform both incentive and matching roles, which often conflict. Performance evaluations are biased because supervisors dislike conflict and are not full residual claimants. CEO pay is strongly related to firm size, which may encourage empire building. The paper's unifying insight is that the people designing and enforcing compensation systems are

usually agents, not true principals, so they may lack incentives to impose efficient contracts on others.

18 Conclusion

This paper is important because it shifts attention from abstract incentive contracts to the messy internal incentive systems of real organizations. The main message is not that incentives do not matter. The message is that incentives matter so much that poorly measured incentives can distort behavior, and organizations therefore rely on imperfect substitutes such as promotions, subjective evaluations, culture, and implicit contracts. The paper also shows that agency problems are layered: the supervisor, CEO, and board members who design incentives are themselves agents with weak and distorted incentives. Understanding compensation practice therefore requires a theory of the firm that combines agency theory, internal labor markets, monitoring incentives, behavioral responses, and organizational culture.